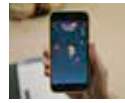




Rajot PUBG ban leads to 10 arrests

Ten youngsters have been arrested for playing PUBG Mobile which has been banned in parts of Gujarat. <https://dgit.in/apr19-45>



Snapchat might get into gaming soon

Snapchat may planning to announce a new gaming platform as early as this month. <https://dgit.in/apr19-46>

Q

uantum computing has become one of the most significant uses of quantum physics.

Our knowledge has

been growing exponentially metamorphosing the computers we use today from abacus in terms of speed and size. But this process is about to reach its physical limitations.

The progression of quantum computers towards the next level will definitely be a stepping stone in modern computing. At the moment companies and researchers are trying to build the most powerful computer the world has ever seen with a handful of different approaches to quantum computing. So, here's everything you need to know about "quantum computing".

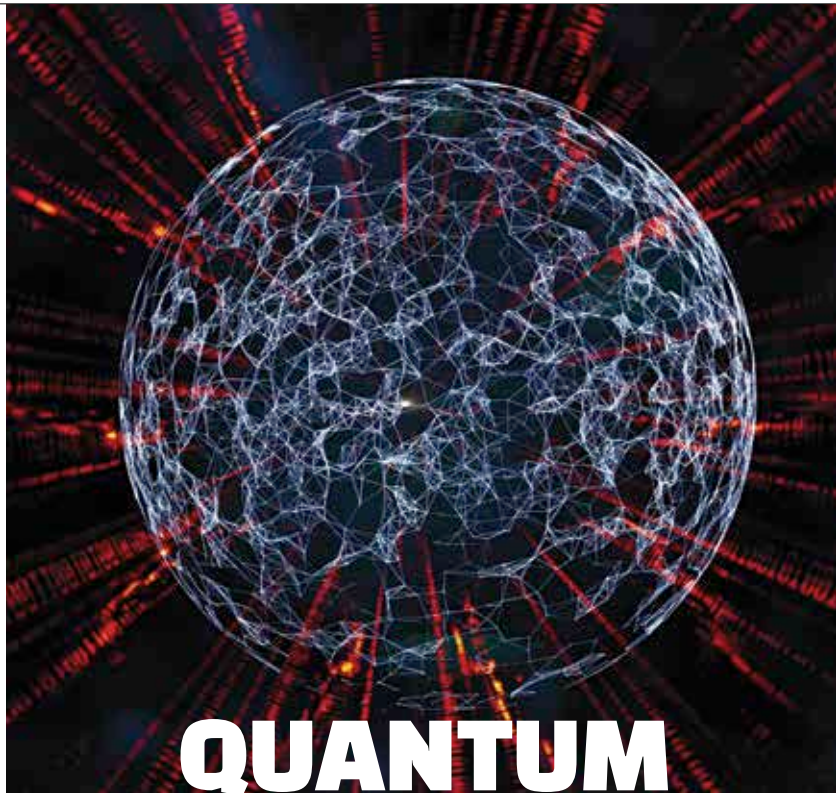
WHAT IS A QUANTUM COMPUTER?

A classical computer performs functions using binary digit or bits and each bit can be a 0 or a 1. Whatever task you want it to perform, it may be booking movie tickets or ordering food the computer takes the input in the form of chains of 1's and 0's, which has been processed by an algorithm. A new string of 1's and 0's gets executed in the end in the monitor in a form of output.

A quantum computer works by taking advantage of the strange ability of electrons, photons and atoms or any of the subatomic particles. Just as a normal computer performs its functions using bits, a quantum computer performs its functions using qubits. Qubits aren't exactly limited to two states. Qubits are considered to be the quantum version of classical binary bits. Qubits make use of two key principles of quantum physics which are "superposition" and "entanglement".

Quantum superposition

In our world, bits exists in either 1's or 0's but in the quantum realm the 1's and 0's can be in any proportions of both states at once. Superposition is defined as the quantum phenomenon which is a consequence of dual particle and wave nature of everything. Erwin



QUANTUM COMPUTING

Exceeding the limits of conventional machines

M. Shreenivasan | feedback@digit.in

Schrödinger imagined taking a cat and placing it in a sealed metal box with a device which had a 50 percent chance of killing the cat. At the end of the experiment he asked what the state of

the cat was. Common sense suggests that the cat may either be alive or dead. But Erwin Schrödinger pointed out that according to quantum theory at the instance before the box was opened the cat was in a "superposition" which means that the cat was neither dead nor alive.

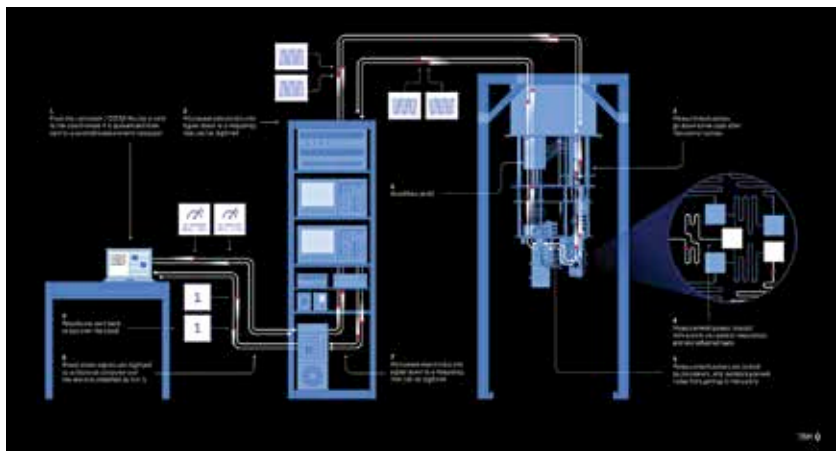


Diagram of a Quantum Computer

Source: IBM Q